习题　5-4

1.计算下列广义积分.

（1）$\int_1^{+\infty}\frac{1}{x^4}dx;$　　（2）$\int_0^{+\infty}e^{-ax}dx(a>0);$

（3）$\int_1^{+\infty}\frac{\arctan x}{x^2}dx;$　　（4）$\int_1^{+\infty}\frac{1}{x^2(1+x)}dx;$

（5）$\int_{-\infty}^{+\infty}\frac{1}{x^2+2x+2}dx;$　　（6）$\int_0^{+\infty}e^{-at}\cos btdt(a>0);$

（7）$\int_0^1\frac{1}{1-x^2}dx;$　　（8）$\int_0^1\sin(\ln x)dx;$

（9）$\int_0^2\frac{1}{x^2-4x+3}dx;$　　(10) $\int_1^e\frac{1}{x\sqrt{1-(\ln x)^2}}dx$

(11) $\int_{\frac{-\pi}{4}}^{\frac{3\pi}{4}}\frac{1}{\cos^2 x}dx;$　　(12) $\int_0^1\frac{1}{\sqrt{1-x^2}}dx;$

(13) $\int_0^{+\infty}x^4e^{-x}dx;$　　(14) $\int_0^{+\infty}x^2e^{-2x^2}dx.$

解（1）$\int_1^{+\infty}\frac{1}{x^4}dx=-\frac{1}{3}\left[x^{-3}\right]_1^{+\infty}=\frac{1}{3};$

（2）$\int_0^{+\infty}e^{-ax}dx=-\frac{1}{a}\left[e^{-ax}\right]_0^{+\infty}=\frac{1}{a};$

（3）$\int_1^{+\infty}\frac{\arctan x}{x^2}\mathrm{d}x=-\int_1^{+\infty}\arctan x\mathrm{d}\left(\frac{1}{x}\right)=-\left[\frac{\arctan x}{x}\right]_1^{+\infty}+\int_1^{+\infty}\frac{1}{x\left(x^2+1\right)}\mathrm{d}x$

$$=\frac{\pi}{4}+\int_1^{+\infty}\left(\frac{1}{x}-\frac{x}{x^2+1}\right)\mathrm{d}x=\frac{\pi}{4}+\left[\ln\frac{x}{\sqrt{x^2+1}}\right]_1^{+\infty}$$

$$=\frac{\pi}{4}+\left(0-\ln\frac{1}{\sqrt{2}}\right)=\frac{\pi}{4}+\frac{1}{2}\ln 2.$$

（4）$\int_1^{+\infty}\frac{1}{x^2(1+x)}\mathrm{d}x=\int_1^{+\infty}\left(-\frac{1}{x}+\frac{1}{x^2}+\frac{1}{1+x}\right)\mathrm{d}x=\left[-\ln x-\frac{1}{x}+\ln\left(1+x\right)\right]_1^{+\infty}$

$$=\left[\ln\left(1+\frac{1}{x}\right)-\frac{1}{x}\right]_1^{+\infty}=0-\left(\ln 2-1\right)=1-\ln 2.$$

（5）$\int_{-\infty}^{+\infty}\frac{1}{x^2+2x+2}\mathrm{d}x=\int_{-\infty}^{+\infty}\frac{1}{\left(x+1\right)^2+1}\mathrm{d}x=\left[\arctan\left(x+1\right)\right]_{-\infty}^{+\infty}$

$$=\frac{\pi}{2}-\left(-\frac{\pi}{2}\right)=\pi.$$

（6） $\int_0^{+\infty}e^{-at}\cos btdt=-\frac{1}{a}\int_0^{+\infty}\cos btde^{-at}=-\frac{1}{a}\left[e^{-at}\cos bt\right]_0^{+\infty}+\frac{1}{a}\int_0^{+\infty}e^{-at}d\cos bt$

$$=\frac{1}{a}-\frac{b}{a}\int_0^{+\infty}e^{-at}\sin btdt=\frac{1}{a}+\frac{b}{a^2}\int_0^{+\infty}\sin btde^{-at}$$
$$=\frac{1}{a}+\frac{b}{a^2}\left[e^{-at}\sin bt\right]_0^{+\infty}-\frac{b}{a^2}\int_0^{+\infty}e^{-at}d\sin bt$$
$$=\frac{1}{a}-\frac{b^2}{a^2}\int_0^{+\infty}e^{-at}\cos btdt.$$

故 $\int_0^{+\infty}e^{-at}\cos btdt=\frac{a}{a^2+b^2}.$

（7） $\int_0^1\frac{1}{1-x^2}\mathrm{d}x=\frac{1}{2}\int_0^1\left(\frac{1}{1-x}+\frac{1}{1+x}\right)\mathrm{d}x=\frac{1}{2}\left[\ln\left|\frac{1+x}{1-x}\right|\right]_0^1=+\infty.$

（8） $\int_0^1\sin(\ln x)\mathrm{d}x\overset{\ln x=t}{=\!=\!=}\int_{-\infty}^0\mathrm{e}^t\sin t\mathrm{d}t=\int_{-\infty}^0\sin t\mathrm{d}\left(\mathrm{e}^t\right)=\left[\mathrm{e}^t\sin t\right]_{-\infty}^0-\int_{-\infty}^0\mathrm{e}^t\cos t\mathrm{d}t$

$$=-\int_{-\infty}^0\mathrm{e}^t\cos t\mathrm{d}t=-\int_{-\infty}^0\cos t\mathrm{d}\left(\mathrm{e}^t\right)=\left[-\mathrm{e}^t\cos t\right]_{-\infty}^0-\int_{-\infty}^0\mathrm{e}^t\sin t\mathrm{d}t$$
$$=-1-\int_{-\infty}^0\mathrm{e}^t\sin t\mathrm{d}t.$$

所以 $\int_0^1\sin(\ln x)\mathrm{d}x=\int_{-\infty}^0\mathrm{e}^t\sin t\mathrm{d}t=-\frac{1}{2}.$

（9） $\int_0^2\frac{1}{x^2-4x+3}\mathrm{d}x=\int_0^1\frac{1}{(x-1)(x-3)}\mathrm{d}x+\int_1^2\frac{1}{(x-1)(x-3)}\mathrm{d}x$

$$=\frac{1}{2}\int_0^1\left(\frac{1}{x-3}-\frac{1}{x-1}\right)\mathrm{d}x+\frac{1}{2}\int_1^2\left(\frac{1}{x-3}-\frac{1}{x-1}\right)\mathrm{d}x.$$

而

$$\int_0^1\left(\frac{1}{x-3}-\frac{1}{x-1}\right)\mathrm{d}x=\frac{1}{2}\left[\ln\left|\frac{x-3}{x-1}\right|\right]_0^1=+\infty,$$

$$\int_1^2\left(\frac{1}{x-3}-\frac{1}{x-1}\right)\mathrm{d}x=\left[\ln\left|\frac{x-3}{x-1}\right|\right]_1^2=-\infty,$$

因右端积分是发散的，故该积分发散.

(10) $\int_1^{\mathrm{e}} \frac{1}{x\sqrt{1-(\ln x)^2}}\mathrm{d}x = \int_1^{\mathrm{e}} \frac{1}{\sqrt{1-(\ln x)^2}}\mathrm{d}(\ln x) = \left[\arcsin(\ln x)\right]_1^{\mathrm{e}} = \frac{\pi}{2} - 0 = \frac{\pi}{2}.$

(11) $\int_{-\frac{\pi}{4}}^{\frac{3\pi}{4}} \frac{1}{\cos^2 x}dx = \int_{-\frac{\pi}{4}}^{\frac{\pi}{2}} \frac{1}{\cos^2 x}dx + \int_{\frac{\pi}{2}}^{\frac{3\pi}{4}} \frac{1}{\cos^2 x}dx = \left[\tan x\right]_{-\frac{\pi}{4}}^{\frac{\pi}{2}} + \left[\tan x\right]_{\frac{\pi}{2}}^{\frac{3\pi}{4}} = \infty$

(12) $\int_0^1 \frac{1}{\sqrt{1-x^2}}dx = \left[\arcsin x\right]_0^1 = \frac{\pi}{2};$

（13） $\int_0^{+\infty} x^4 \mathrm{e}^{-x}\mathrm{d}x = \Gamma(5) = 4! = 24.$

（14） $\int_0^{+\infty} x^2 \mathrm{e}^{-2x^2}\mathrm{d}x \overset{2x^2=t}{=\!=\!=} \int_0^{+\infty} \frac{t}{2}\mathrm{e}^{-t}\mathrm{d}\left(\sqrt{t}\right) = \frac{1}{4\sqrt{2}}\int_0^{+\infty} t^{\frac{1}{2}}\mathrm{e}^{-t}\mathrm{d}t$

$$= \frac{1}{4\sqrt{2}}\int_0^{+\infty} t^{\frac{3}{2}-1}\mathrm{e}^{-t}\mathrm{d}t = \frac{1}{4\sqrt{2}}\Gamma\left(\frac{3}{2}\right) = \frac{1}{4\sqrt{2}}\cdot\frac{1}{2}\Gamma\left(\frac{1}{2}\right) = \frac{1}{8}\sqrt{\frac{\pi}{2}}.$$

2.求$\Gamma(\frac{11}{2})$的值.

解 $\Gamma(\frac{11}{2}) = \Gamma(\frac{9}{2}+1) = \frac{9}{2}\Gamma(\frac{9}{2}) = \frac{9}{2}\Gamma(\frac{7}{2}+1) = \frac{9}{2}\cdot\frac{7}{2}\Gamma(\frac{7}{2}) = \cdots$

$$= \frac{9}{2}\cdot\frac{7}{2}\cdot\frac{5}{2}\cdot\frac{3}{2}\cdot\frac{1}{2}\Gamma(\frac{1}{2}) = \frac{9}{2}\cdot\frac{7}{2}\cdot\frac{5}{2}\cdot\frac{3}{2}\cdot\frac{1}{2}\sqrt{\pi} = \frac{945}{32}\sqrt{\pi}.$$

3.计算广义积分$\int_0^{+\infty} x^3 e^{-\lambda x}dx$，其中$\lambda$为大于零的常数.

解 $\int_0^{+\infty} x^3 e^{-\lambda x}dx = \frac{1}{\lambda}\int_0^{+\infty} x^3 e^{-\lambda x}d(\lambda x) \overset{\lambda x=t}{=\!=\!=} \frac{1}{\lambda^4}\int_0^{+\infty} t^3 e^{-t}dt = \frac{1}{\lambda^4}\Gamma(4) = \frac{3!}{\lambda^4} = \frac{6}{\lambda^4}.$

4.当k这何值时，广义积分$\int_2^{+\infty} \frac{1}{x(\ln x)^k}dx$收敛？当$k$为何值时，广义积分发散？

解 由$\int \frac{\mathrm{d}x}{x(\ln x)^k} = \int \frac{\mathrm{d}(\ln x)}{(\ln x)^k} = \begin{cases} \ln\ln x + C, & k = 1, \\ \frac{1}{1-k}(\ln x)^{-k+1} + C, & k \neq 1, \end{cases}$

故

当$k<1$时，$\int_2^{+\infty} \frac{\mathrm{d}x}{x(\ln x)^k} = \left[\frac{1}{1-k}(\ln x)^{-k+1}\right]_2^{+\infty} = +\infty$；

当$k=1$时，$\int_2^{+\infty} \frac{\mathrm{d}x}{x(\ln x)^k} = \left[\ln(\ln x)\right]_2^{+\infty} = +\infty$；

当$k>1$时，$\int_2^{+\infty}\frac{\mathrm{d}x}{x(\ln x)^k}=\left[\frac{1}{1-k}(\ln x)^{-k+1}\right]_2^{+\infty}=\frac{1}{k-1}(\ln 2)^{1-k}$.

因此，当$k>1$时，广义积分收敛； 当$k\le 1$时，广义积分发散.